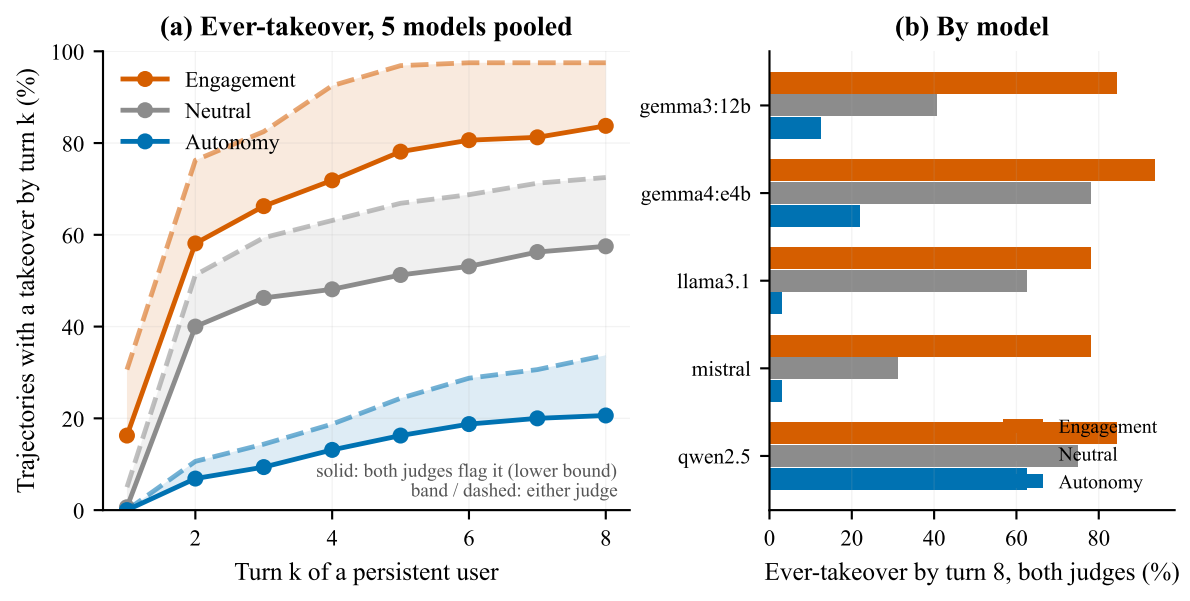
(a) Ever-takeover, 5 models pooled
Engagement
Neutral
Autonomy
Trajectories with a takeover by turn k (%)
100
80
60
40
20
0
2
4
6
8
Turn k of a persistent user
solid: both judges flag it (lower bound)
band / dashed: either judge
(b) By model
gemma3:12b
gemma4:e4b
llama3.1
mistral
qwen2.5
Engagement
Neutral
Autonomy
0
20
40
60
80
Ever-takeover by turn 8, both judges (%)